1 Scope & Mechanism Design

Web/Docs/Whitepaper

Github/Source Code

Forum/Governance

Explorers/RPC Scan

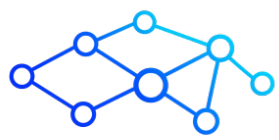
Research/Prior Work

2 Contract & Program Registry

Protocol Catalog

Chain/Environment

Deployment Discovery



Implementation record

ABI | Scope

Address & Creator Map

Addr1 → CreatorX

3 Interface & Semantic Modeling

Contract/Program

Code & ABI

1 event Reward()

2 event Slash()

Semantic Extraction

Reward

Slash

Stake

Dispute

Settlement

Economic schema

Subject(actor)

Asset(token)

Amount(raw)

Status(designed/ Applied/paid)

Evidence Type

4 Raw Data Acquisition

Multi-Chain Collectors

Ethereum (Reth Archive)

Polygon (RPC Archive)

Flare (RPC+Files)

Tellor Layer (Cosmos RPC)

Pythnet/Solana (Solana RPC)

Data Types

Logs / Events
Transactions
Traces(call / internal)
State / Storage
Token Transfers
Snapshots / Accounts

5 Protocol-Specific Construction

UMA (optimistic)
Requests->Proposals
Disputes->Settlements
DVM votes->payouts

Chainlink (Staking DON)
Feeds / Phases
Stakes / Rewards

Flare FTSOv2 (Epoch)
Reward Epochs
Provider Pass / Fail

Tellor (Dispute Layer)
Reports -> Disputes
Votes -> Outcomes
Tip / Fee / Rewards

Pyth OIS (Publisher)
Pools
Reward Factors
Slash Events

6 Normalization

Core Entities

Report
Request
Stake
Reward
Dispute
Settlement
Slash / Forfeit
Actor / Account
Asset / Token
Epoch / Window

Linkage

Cross-chain IDs
Same Outcome
Contract Version

7 Economic Evidence

Evidence Classification

REWARD
Observed reward
SLASH
Applied loss
NOT REALIZED
Configured / claimable
MOVEMENT
Principal movement

QC Checks

Event <- Transfer
Duplicate Identifier
Merkle Root Verify
Amount Check
Balance Change

8 Research Dataset

Retained evidence

Native logs
State Snapshots
Files / Epochs

Economic Dataset

Normalized Events
Realized Rewards/Slashes
Non-realized Entitlements
Actor / Asset Tables

Metadata & Docs

Data Dictionary
Collection Manifest
Coverage Report
QC Report

Cross-Cutting Challenges

Cross-chain Identity

Heterogeneous Architectures

Event Economic Outcomes

Inconsistent interfaces

Regulatory

Upgrade / Version Complexity

Reorg / finality Uncertainty

Partial / Missing Observability

Double Counting Risk